

34721/34722 Linear Control Design 1

Spring Semester

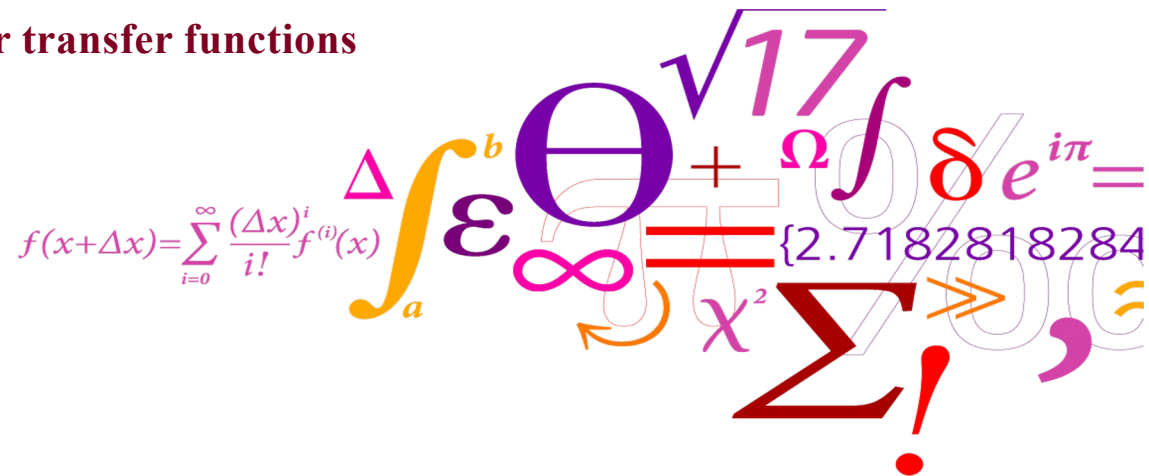
Silvia Tolu - Dimitrios Papageorgiou



Lecture 5/13

Schedule (15:00-17:00)

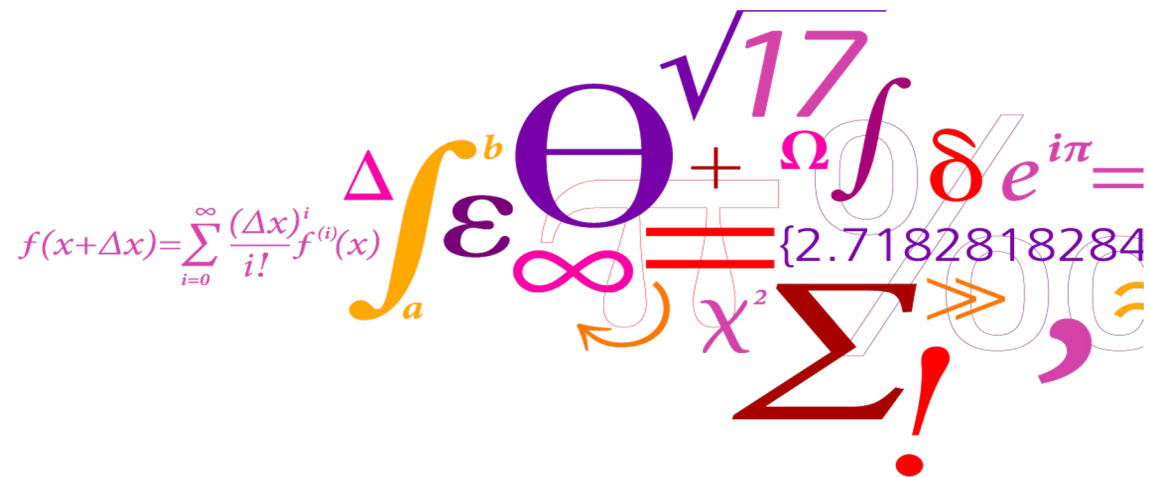
- **Black-box modelling**
- **From ODEs to transfer functions**
- **Linearisation of nonlinear ODEs for transfer functions**
- **Exercises**



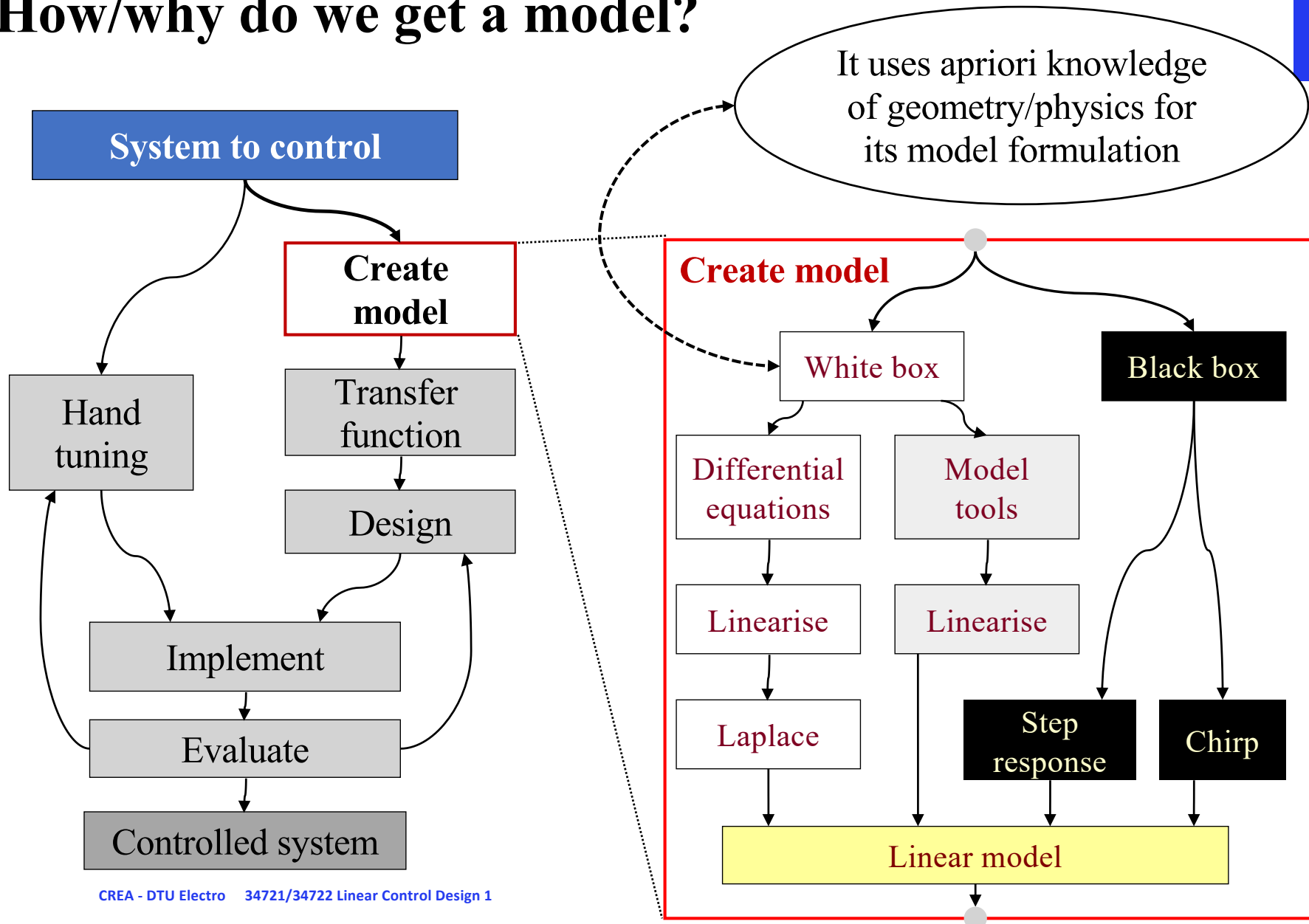
What you will learn today

After this lecture, you will be able to

- Understand the important role of **modelling** in the control system design process
- Explain the different types of modelling, i.e., **white, grey, and black box modelling**
- **Identify transfer functions** of systems from input/output data
- **Linearise a nonlinear model** of a physical system to obtain its transfer function



How/why do we get a model?



Approach to Dynamic System Modelling

1. Define the system and its components
2. Formulate the mathematical model, express fundamental necessary **assumptions**
3. Obtain **differential equations** representing the mathematical model (**laws of physics**)
4. Linearise the system, apply **Laplace Transform** and find the **transfer functions**
5. Test the model with real data and assess the validity of the assumptions
6. If necessary, re-analyse or re-model the system

Modelling

We will only work with **linear** systems



Modelling can be done in 3 ways:

➤ White-box modelling

- known equations (from laws of physics or empirical)
- known parameters (e.g. from datasheets)

$$10\ddot{x} = 5.2\dot{x} + 7x + 4u$$

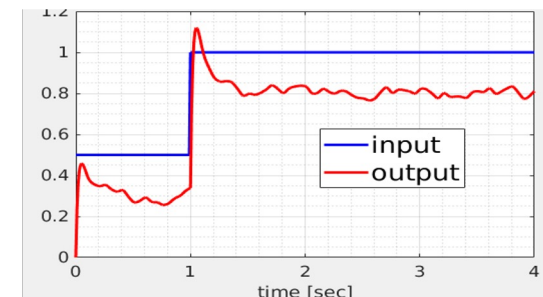
➤ Grey-box modelling

- known equations (from laws of physics or empirical)
- unknown parameters (need to be determined)

$$m\ddot{x} = b\dot{x} + kx + cu$$

➤ Black-box modelling

- unknown equations (model structure needs to be decided)
- unknown parameters (need to be determined from experiments)



Black-Box Modelling

We will only work with **linear** systems



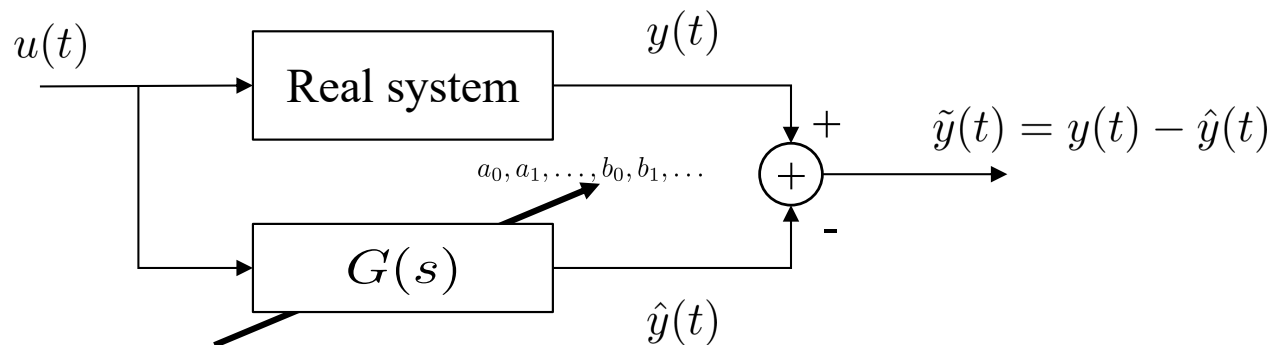
Obtaining a model from just measurements involves two steps:

1. **Choose a model structure** → i.e., how the transfer function polynomials look, for example

$$G(s) \triangleq \frac{y(s)}{u(s)} = \frac{b}{s+a}, \text{ or } \frac{b_1s+b_0}{s^2+a_1s+a_0}, \text{ or } \frac{b_3s^3+b_2s^2+b_1s+b_0}{s^3+a_2s^2+a_1s+a_0}$$

2. **Obtain reasonable values for the coefficients** of these polynomials

- Not a trivial task → needs much data and many tests.
- This is called “*System Identification*” (more on that in MSc).
- **Main idea:** Compare the simulated output with the real output and adjust parameters until the compared outputs match.



Adjust $a_0, a_1, \dots, b_0, b_1$ such that the error between real and simulated output is minimised.

Black-Box Modelling

Given time series for input $u(t)$ and output $y(t)$

Is the model static (no transfer functions)?

- We can fit a static relation between u and y , e.g., $y(t) = 4.5x^2(t)$

Is the system dynamical (derivatives, etc.)?

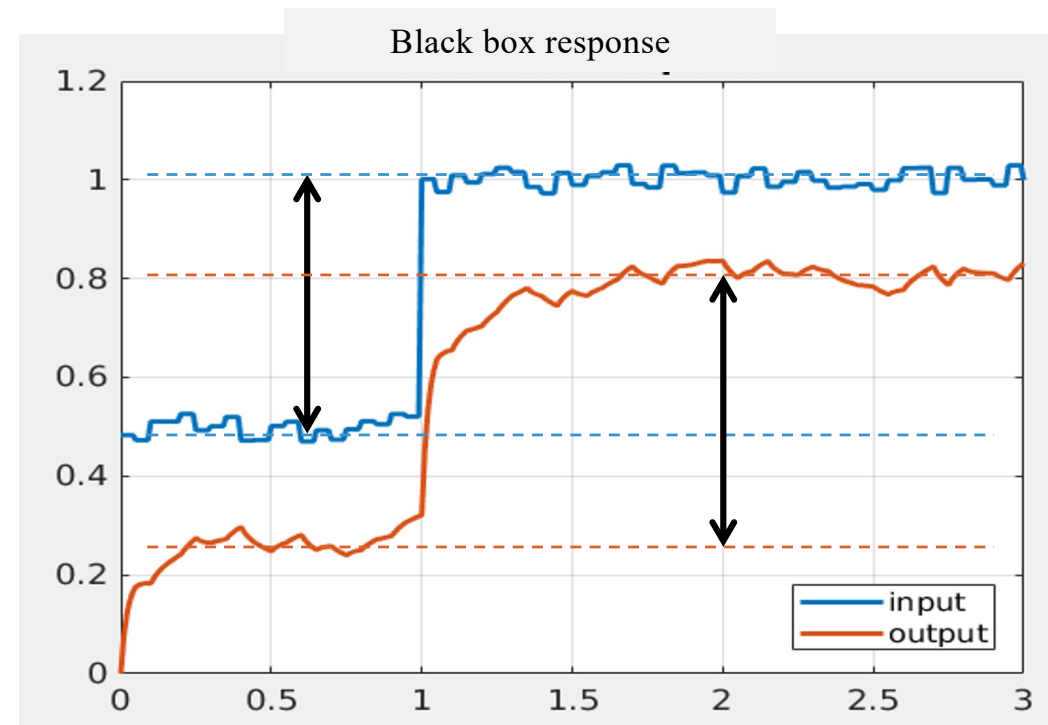
- We can fit a transfer function

What is the transfer function?

- We can start from the steady-state gain

In the steady state (when everything has settled), we can see that

$$K_{SS} \triangleq \frac{y_{SS}}{u_{SS}} = \frac{0.8 - 0.3}{1 - 0.5} = 1$$



The system is dynamical → We need to check for poles and zeros.

Black-Box Modelling

Given time series for input $u(t)$ and output $y(t)$

Is the system dynamical (derivatives, etc.)?

- Perhaps we can fit a transfer function

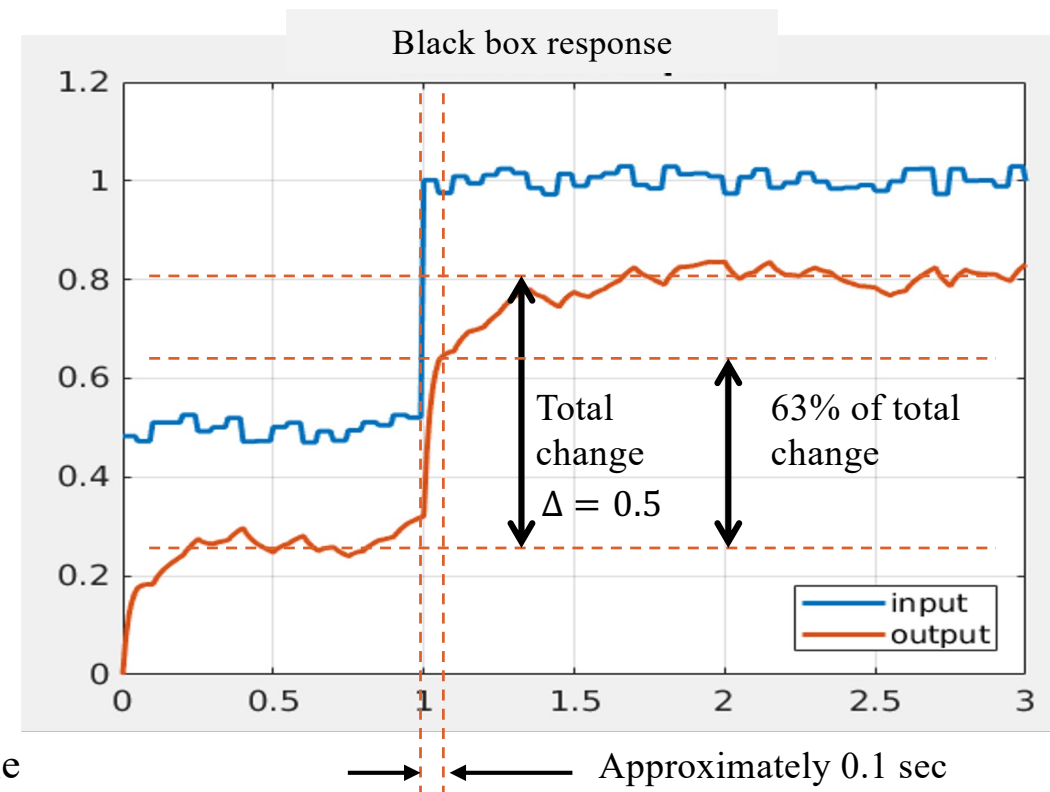
How many poles and zeros?

- Choose model** (polynomials) **structure**, e.g., 1 pole and no zeros

$$G(s) \triangleq \frac{Y(s)}{U(s)} = \frac{b_0}{a_1 s + 1}$$

$$K_{SS} = \lim_{s \rightarrow 0} G(s) = \frac{b_0}{1} = 1 \Rightarrow b_0 = 1$$

The parameter a_1 is the **time constant** τ_b , i.e., the time it takes for the output to go from standstill to 63% of its **total change** → from 0.3 to $0.63 \cdot 0.5 \cong 0.315$ (measured from 0.3, so it is 0.615). This is approx. 0.1 sec → $a_1 = 0.1$



$$G(s) = \frac{1}{0.1s + 1}$$

Black-Box Modelling



Given time series for input $u(t)$ and output $y(t)$

What if we have more poles and zeros?

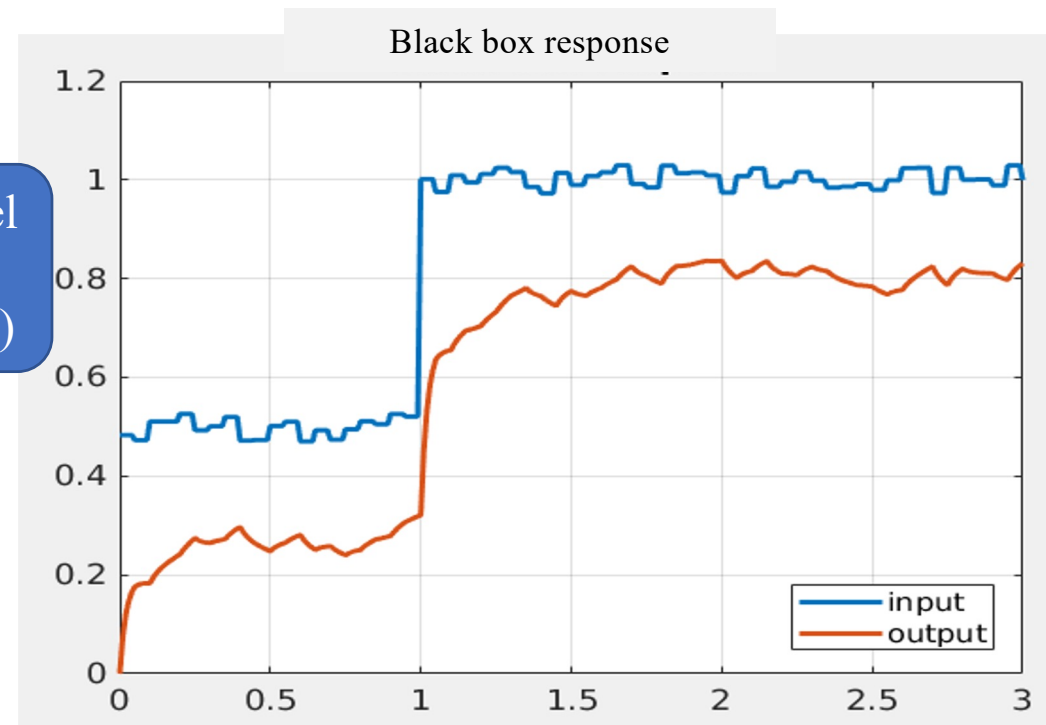
$$G(s) = \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + 1}$$

$$G(s) = \frac{b_0}{a_3 s^3 + a_2 s^2 + a_1 s + 1}$$

Choose model
structure
(polynomials)

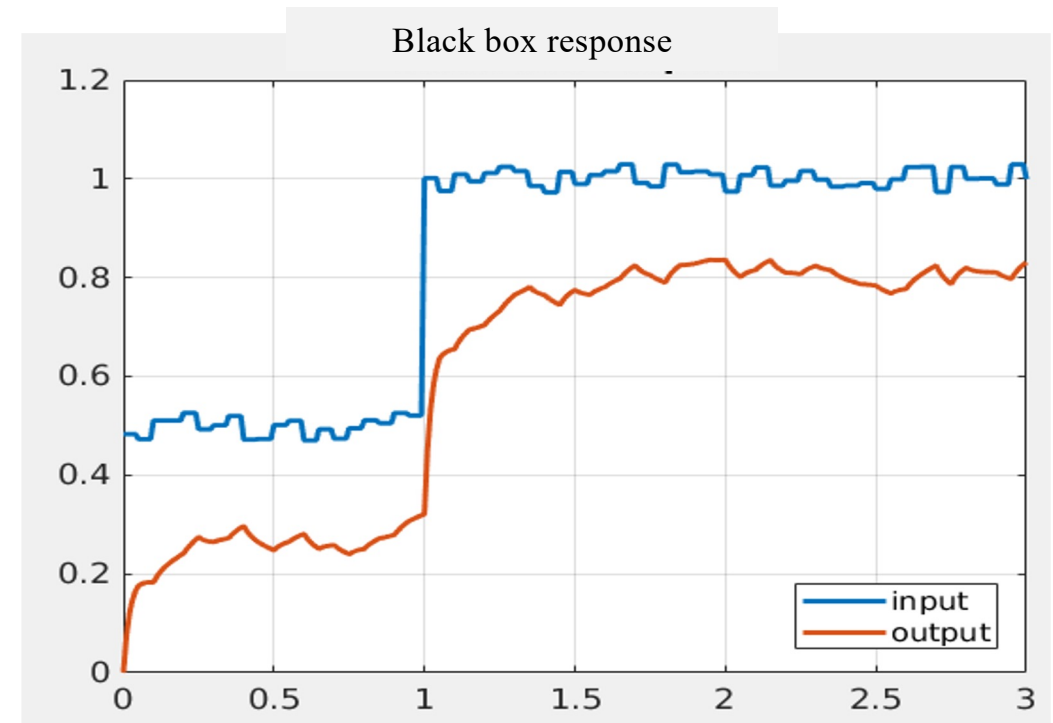
There may be too many parameters, but Matlab can help ☺

→ **System Identification Toolbox**



Black-Box Modelling – Matlab SysId Toolbox

1. We make experiments and record at least a pair of input-output time series
2. We choose a model structure, i.e., we decide how many poles and zeros we will have in our transfer function → iterative process
3. We solve an optimisation problem to identify the transfer function polynomial coefficients
4. **The goal is to have the smallest possible error between the simulated (based on the identified transfer function) output and the recorded real output**



Black-Box Modelling – Matlab SysId Toolbox



```
clear all;
clc;

data = readtable('datafile1.txt'); % Load data file
t = table2array(data(:,1)); % First column is time
u = table2array(data(:,2)); % Second column is input
y = table2array(data(:,3)); % Third column is output

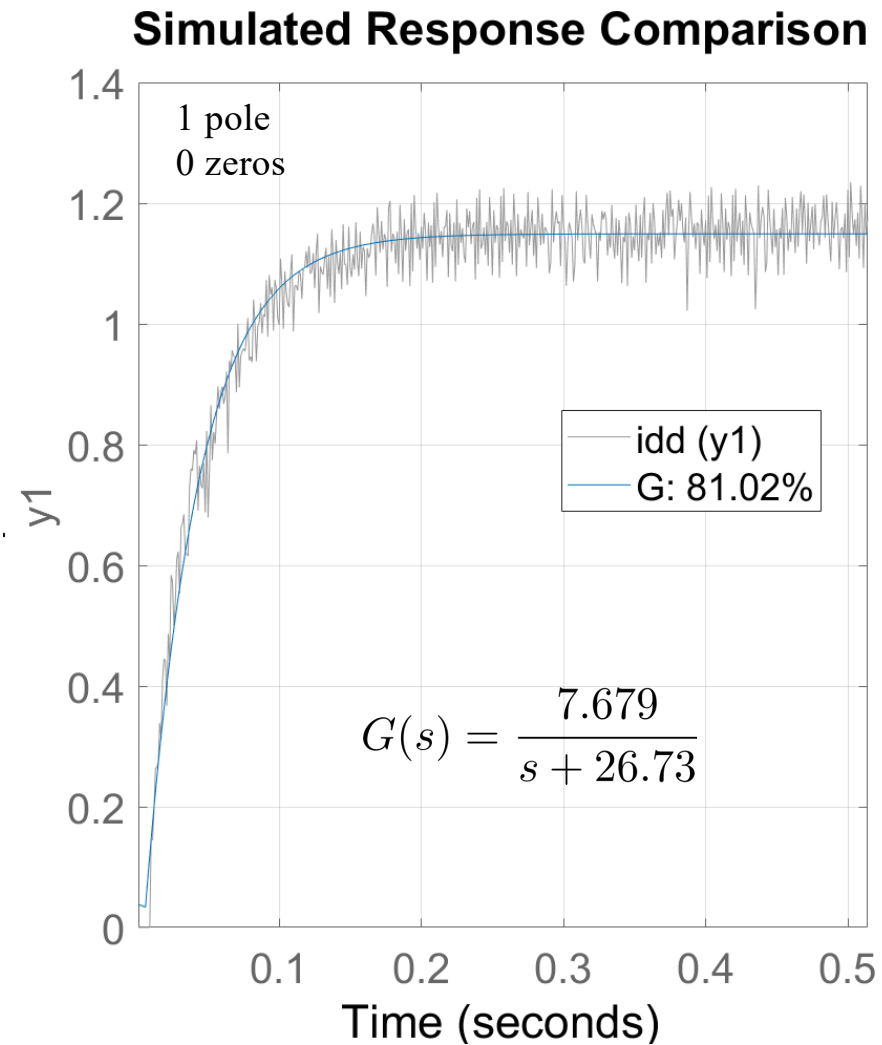
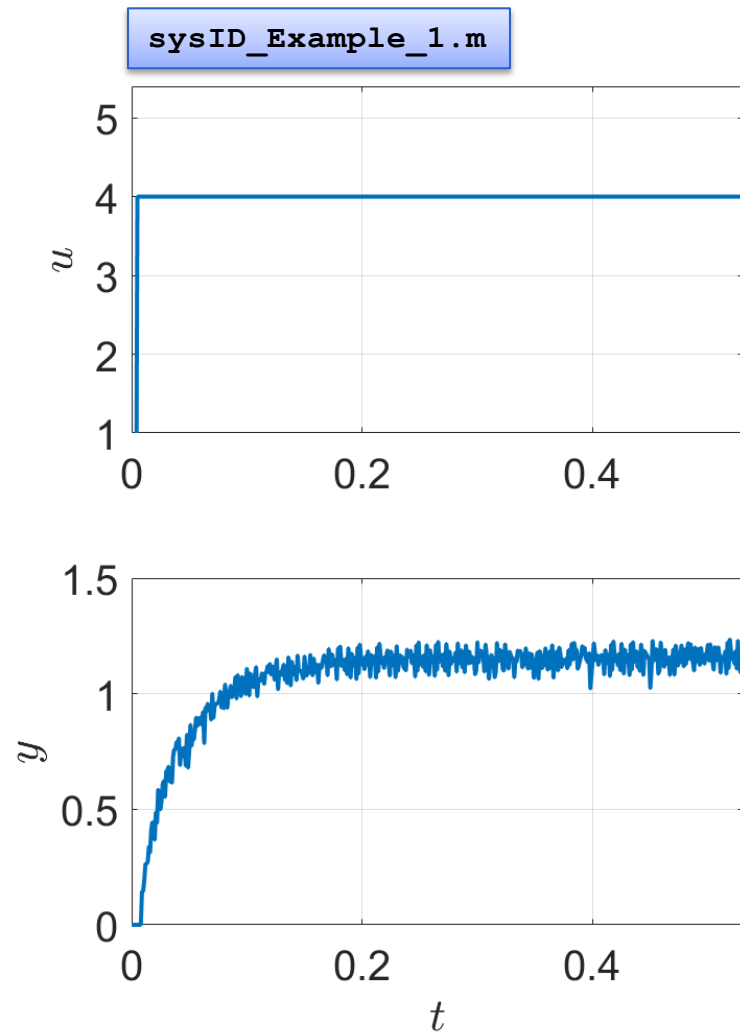
T_s = t(2) - t(1); % Calculate the Sampling time
idd = iddata(y,u,T_s); % Put inputs and outputs to special structure
np = 1; % Select number of poles
nz = 0; % Select numbers of zeros

G = tfest(idd,np,nz); % Identify the transfer function coefficients
step(G); % Make a step response plot of G

figure(2);
compare(G,idd); % Compare and plot real and simulated outputs
grid on;
```

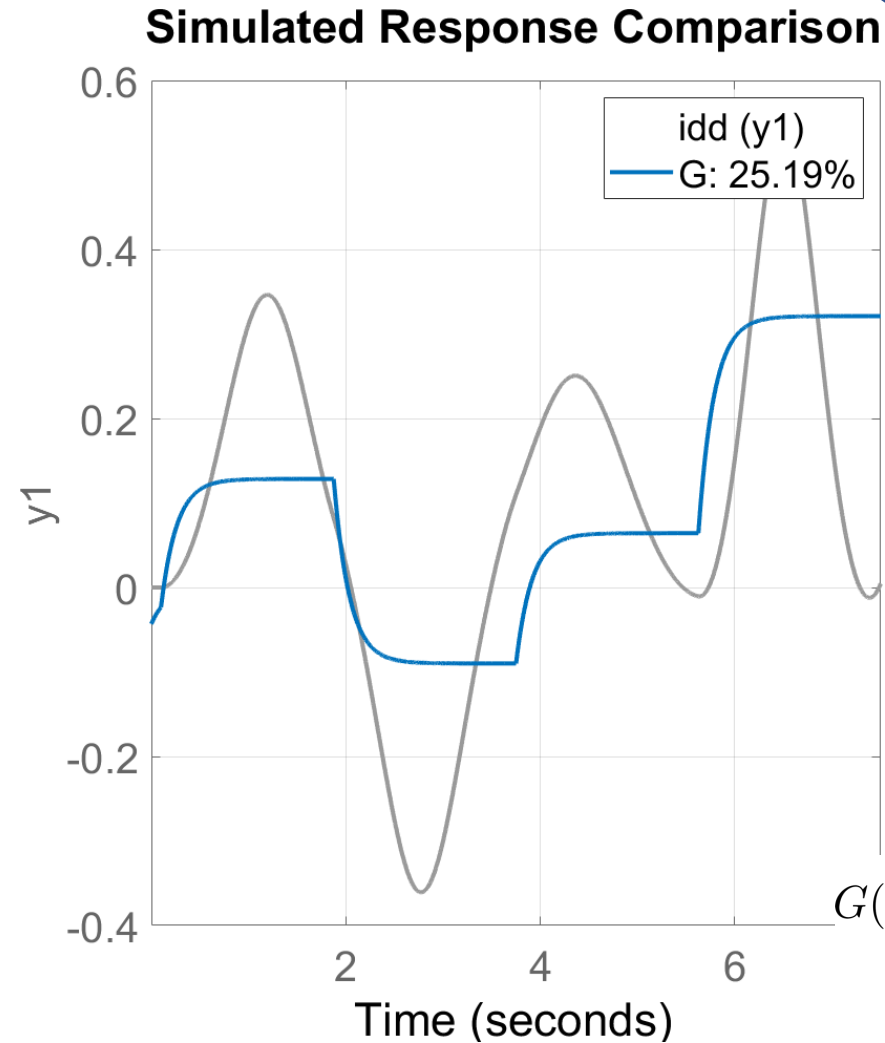
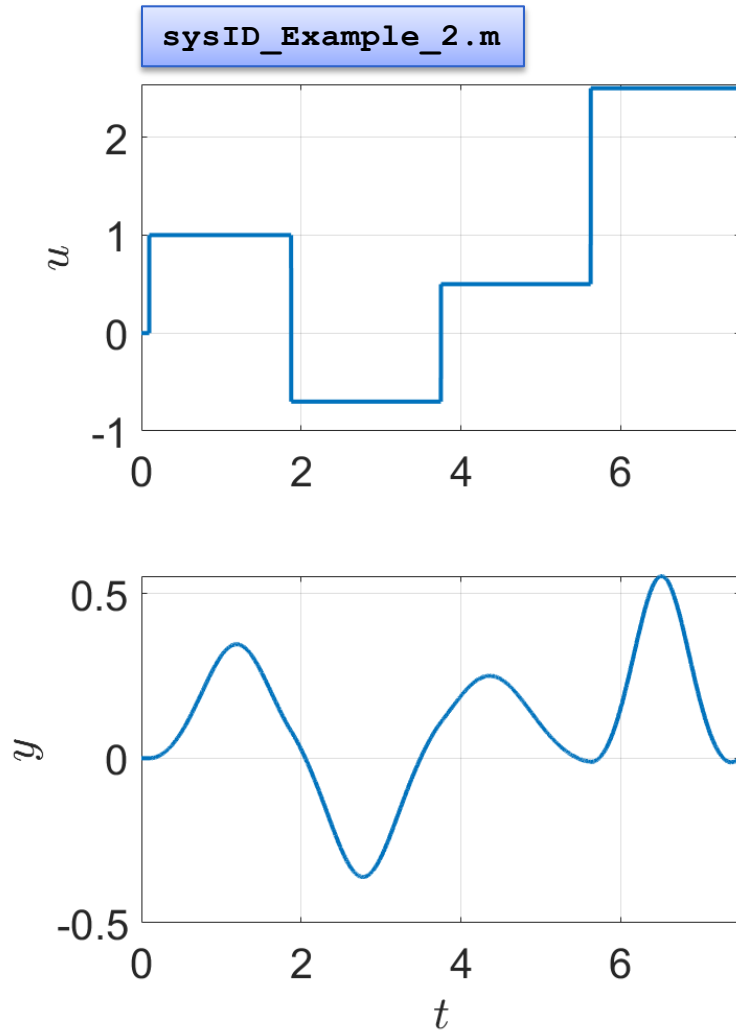
sysID_Example_1.m

Black-Box Modelling – Matlab SysId Toolbox



Black-Box Modelling – Matlab SysId Toolbox

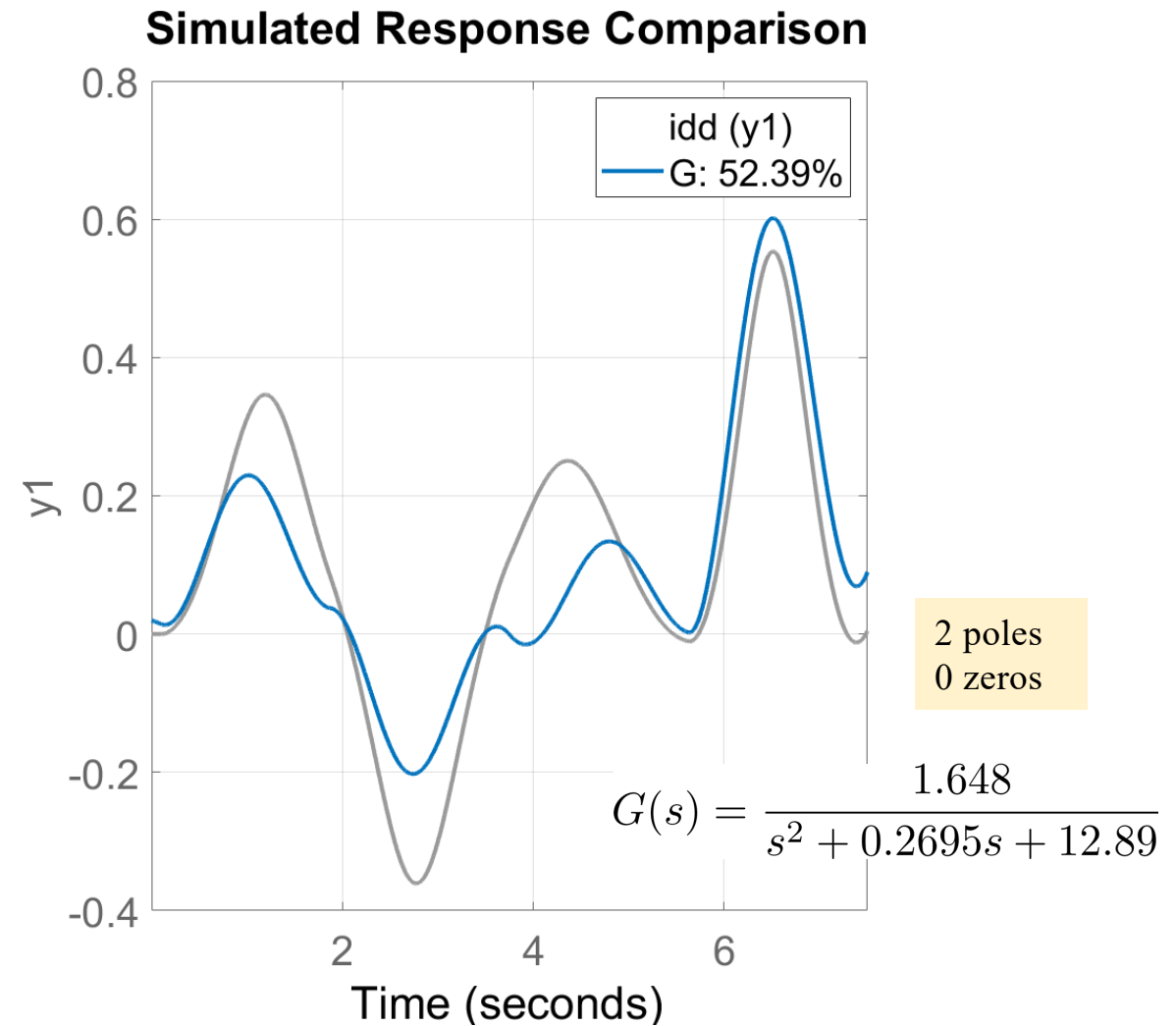
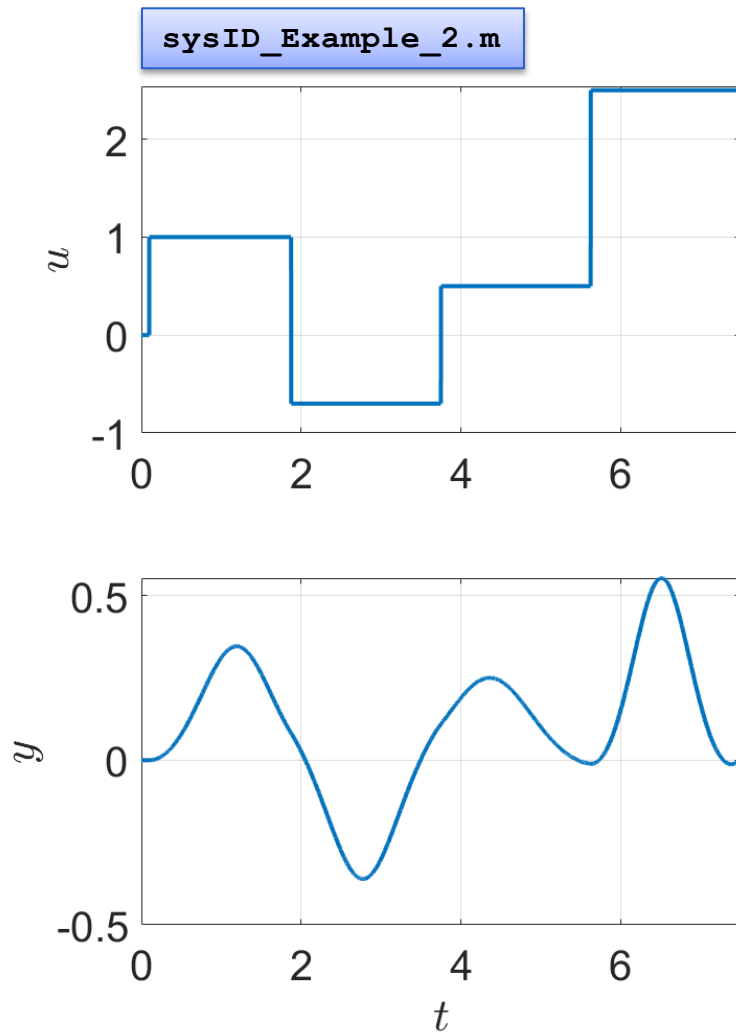
Find the best set of parameters to minimize the output error



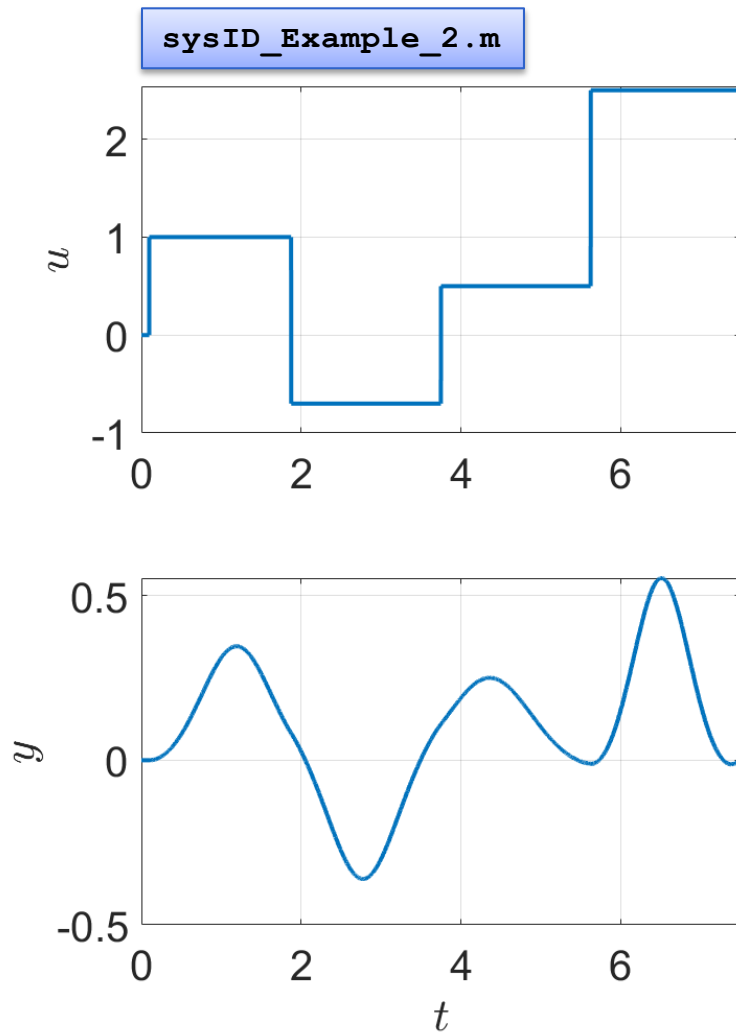
1 pole
0 zeros

$$G(s) = \frac{0.7989}{s + 6.22}$$

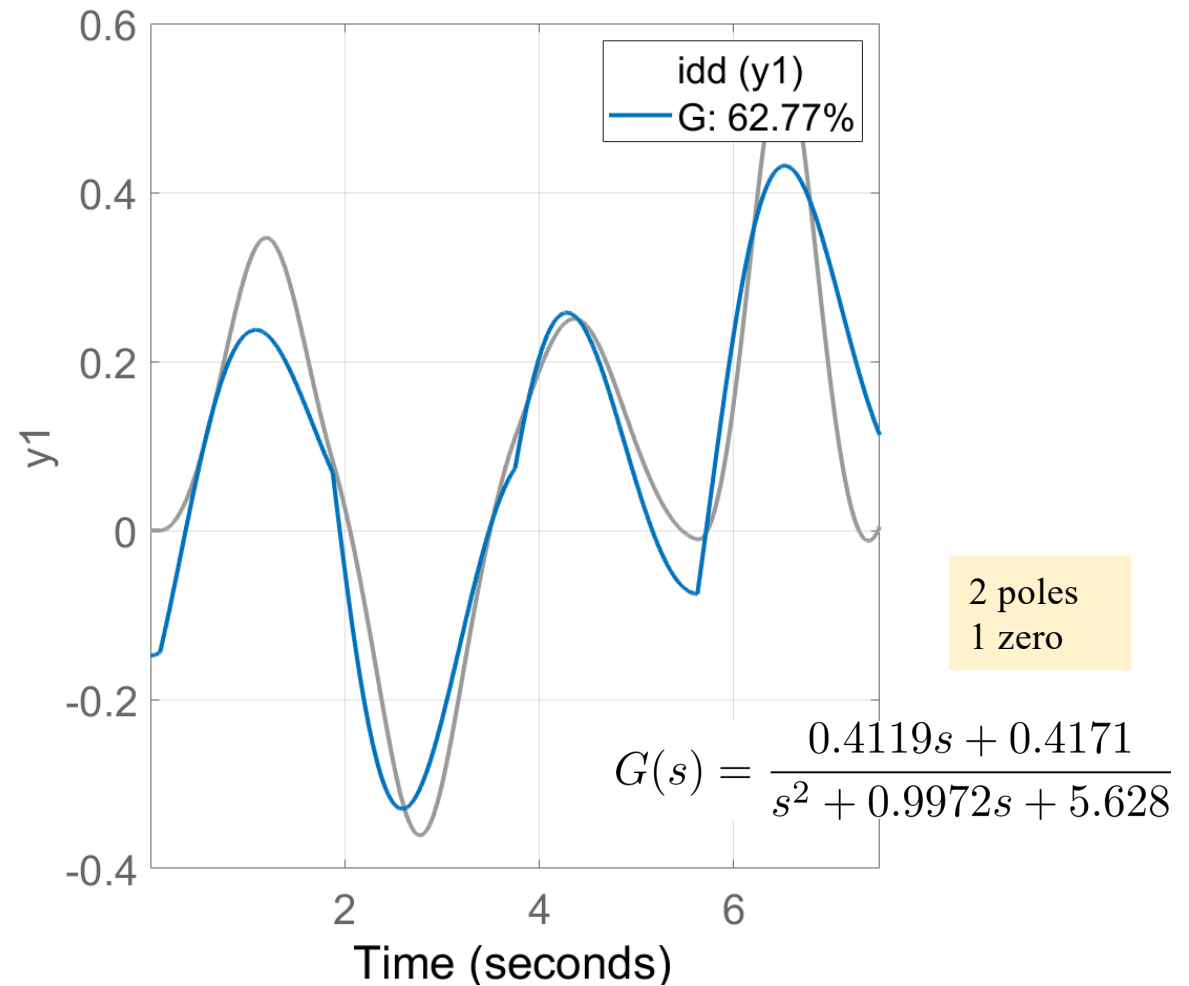
Black-Box Modelling – Matlab SysId Toolbox



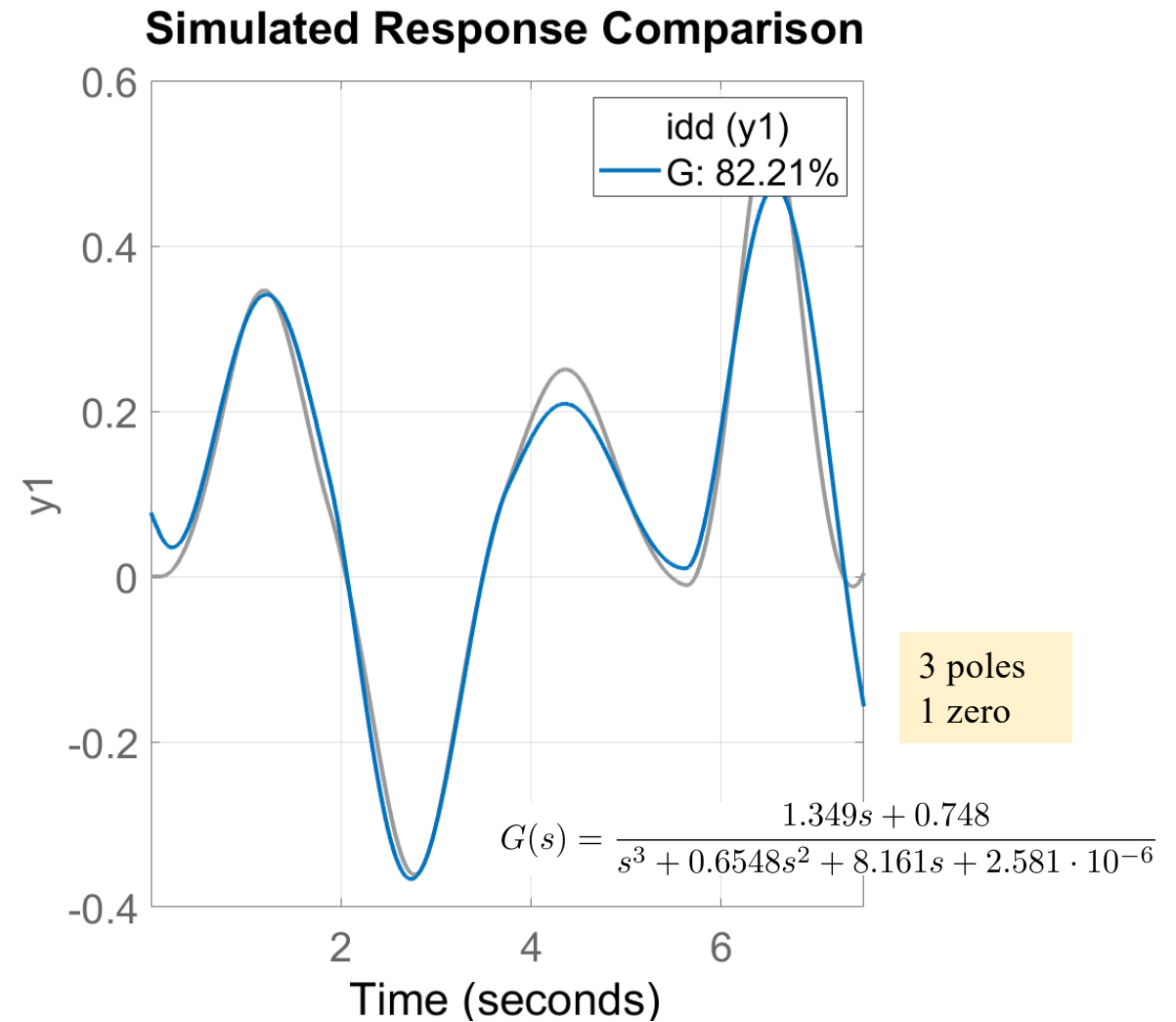
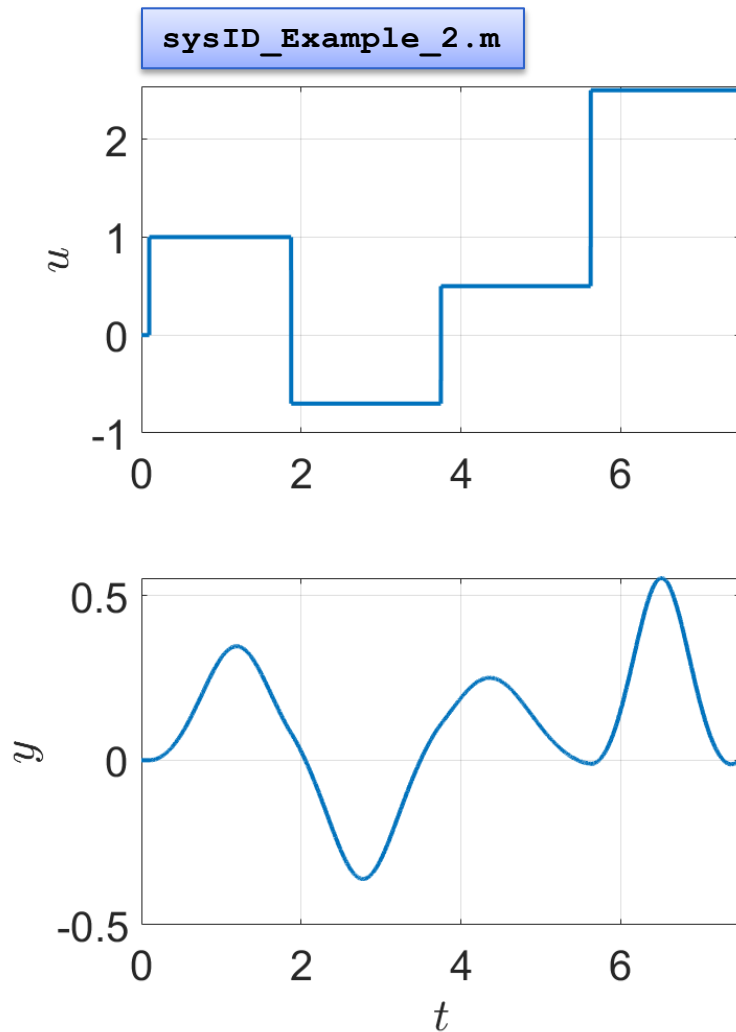
Black-Box Modelling – Matlab SysId Toolbox



Simulated Response Comparison



Black-Box Modelling – Matlab SysId Toolbox

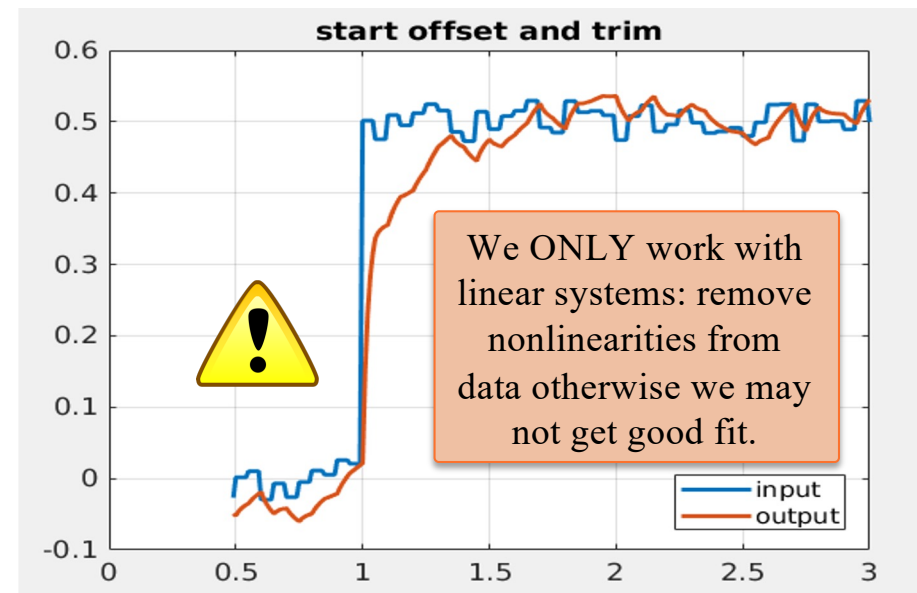
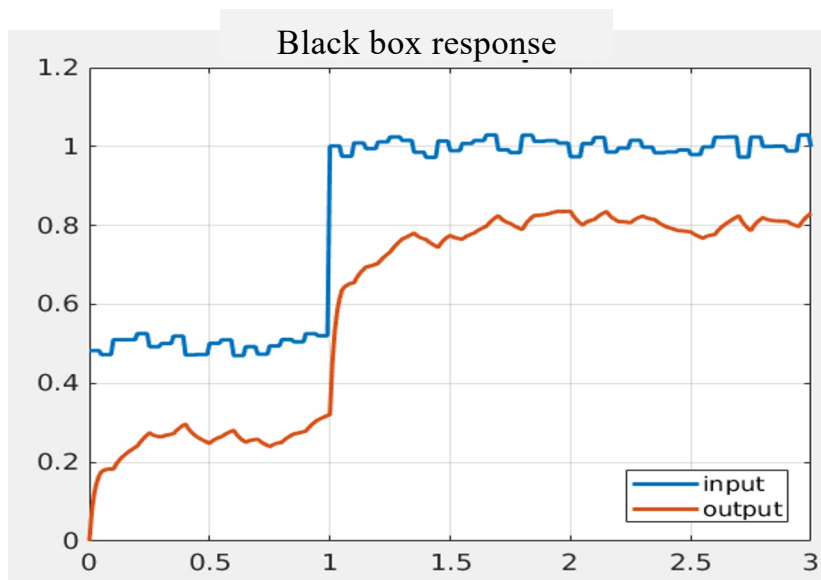


Black-Box Modelling – Matlab SysId Toolbox

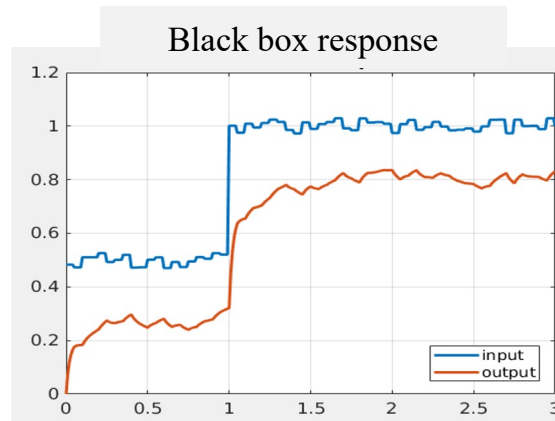
We are looking **for a linear model**. A linear (stable) system with 0 in input gives 0 in output.

The system below is **non-linear** - only change at 1 sec

The start (up until 0.5 sec) must be omitted →→ Offset data so they start at 0

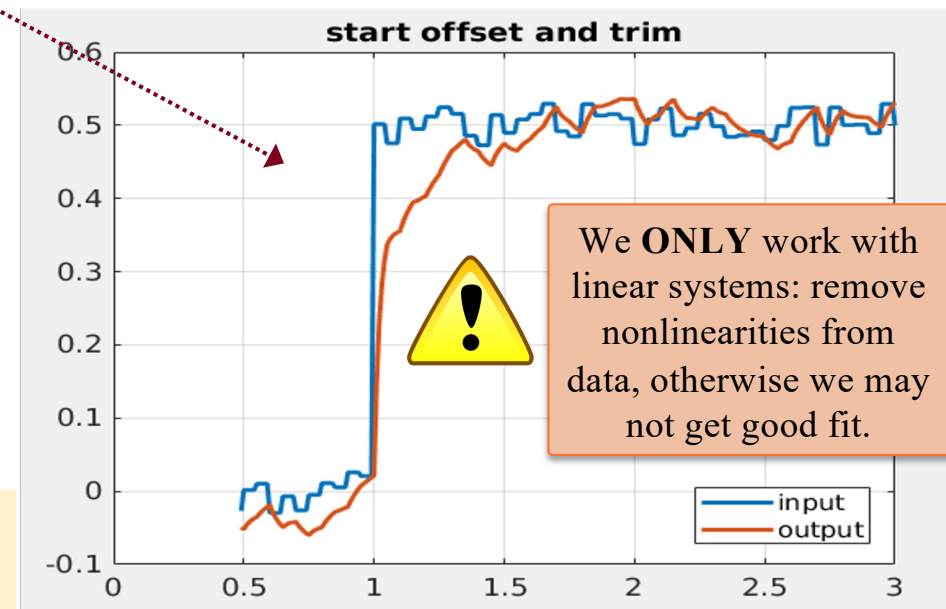


Black-Box Modelling - MATLAB Transfer Function Estimate



```
data = load('fil.txt');  
dd = iddata( ...  
    data(50:end, 3) - offset_output, ...  
    data(50:end, 2) - offset_input, ...  
    0.01);
```

We start with data around 0
We cut data before 0.5 sec
Offset output is 0.3 circa
Offset input is 0.5 circa
Optional. Cut the data where there is a lot of noise
(between 2 and 3 sec)



Black-Box Modelling

MATLAB Transfer Function Estimate



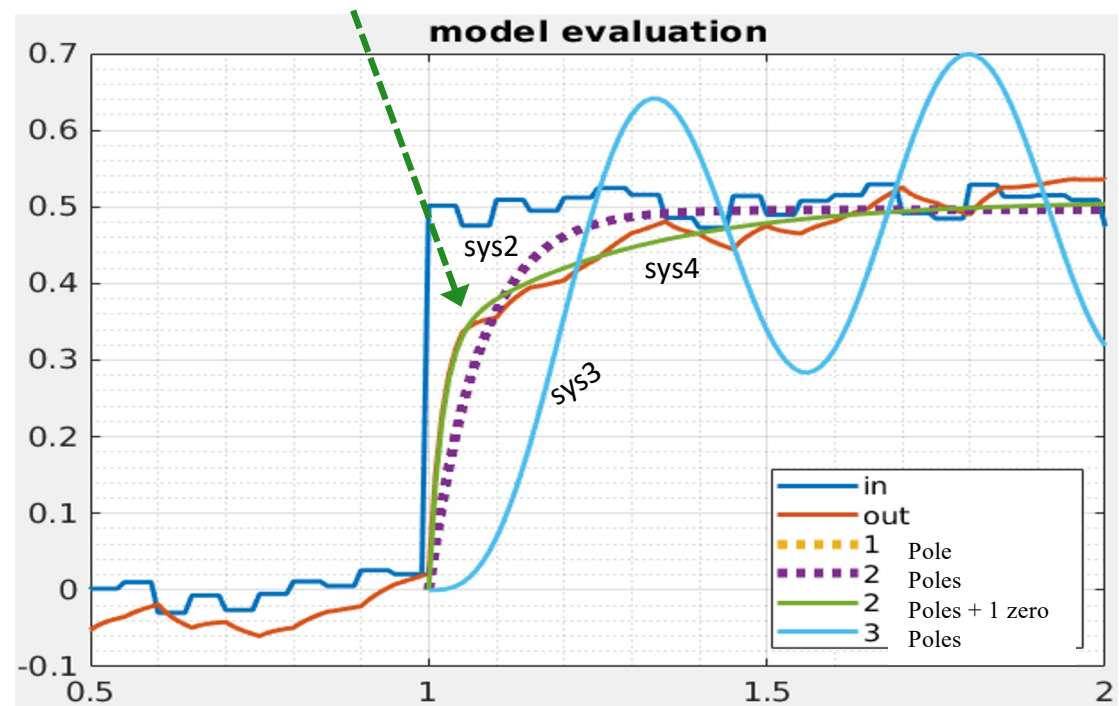
```
data = load('fil.txt');  
dd = iddata( ...  
    data(50:end,3) - offset_output, ...  
    data(50:end,2) - offset_input, ...  
    0.01);
```

```
sys2 = tfest(dd, 2, 0)  
sys3 = tfest(dd, 3, 0)  
sys4 = tfest(dd, 2, 1)
```

We start in 0.5 sec. and we finish in 2 sec.
We re-do the TF estimation.

SYS2 (2 poles purple). It takes longer and overshoots.
SYS3 (3 poles blue). It is not good at all.
SYS4 (2 poles and 1 zero, green) fits well!

with a clean step from 0.5 to 1 sec



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From ODEs to transfer functions

Dynamical systems (unless they are identified as a black-box from data) are modelled by Ordinary Differential Equations (ODEs)

$$m\ddot{x} = u - \beta\dot{x} - kx$$

Mass-spring-damper

$$\ddot{\theta} = \frac{1}{ml^2}u - \frac{k}{m}\dot{\theta} - \frac{g}{l}\sin\theta$$

Pendulum

$$\dot{h} = \frac{1}{A}(\alpha u - k\sqrt{h} - q_{\text{leak}})$$

Single water tank

$$m\dot{v} = u - b|v|v$$

Ship surge velocity

Linear ODEs give transfer functions with a simple operation. This is not the case with nonlinear systems

$$m\ddot{x} = u - \beta\dot{x} - kx \xrightarrow{\mathcal{L}}$$

$$ms^2X(s) = U(s) - \beta sX(s) - kX(s) \Leftrightarrow$$

$$(ms^2 + \beta s + k)X(s) = U(s) \Leftrightarrow$$

$$G(s) \triangleq \frac{X(s)}{U(s)} = \frac{1}{ms^2 + \beta s + k}$$

From ODEs to transfer functions

- A dynamical system (unless identified as a black-box from data) is modelled by Ordinary Differential Equations (ODEs).
- Linear systems (linear ODEs) can always be expressed in the Laplace (frequency) domain.
- This allows working with algebraic (polynomial) expressions instead of ODEs (more difficult).
- If the systems (ODEs) are **nonlinear**, e.g., $m\dot{v} = u - b|v|v$, then we need to **linearise** them first using the Taylor series expansion.



A continuous function $f(x)$ can be expanded around a point x_0 of its domain as

$$f(x) = f(x_0) + \underbrace{\frac{f'(x_0)}{1!}(x - x_0)}_{\text{linear term}} + \underbrace{\frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \dots}_{\text{higher order terms}}$$

From ODEs to transfer functions

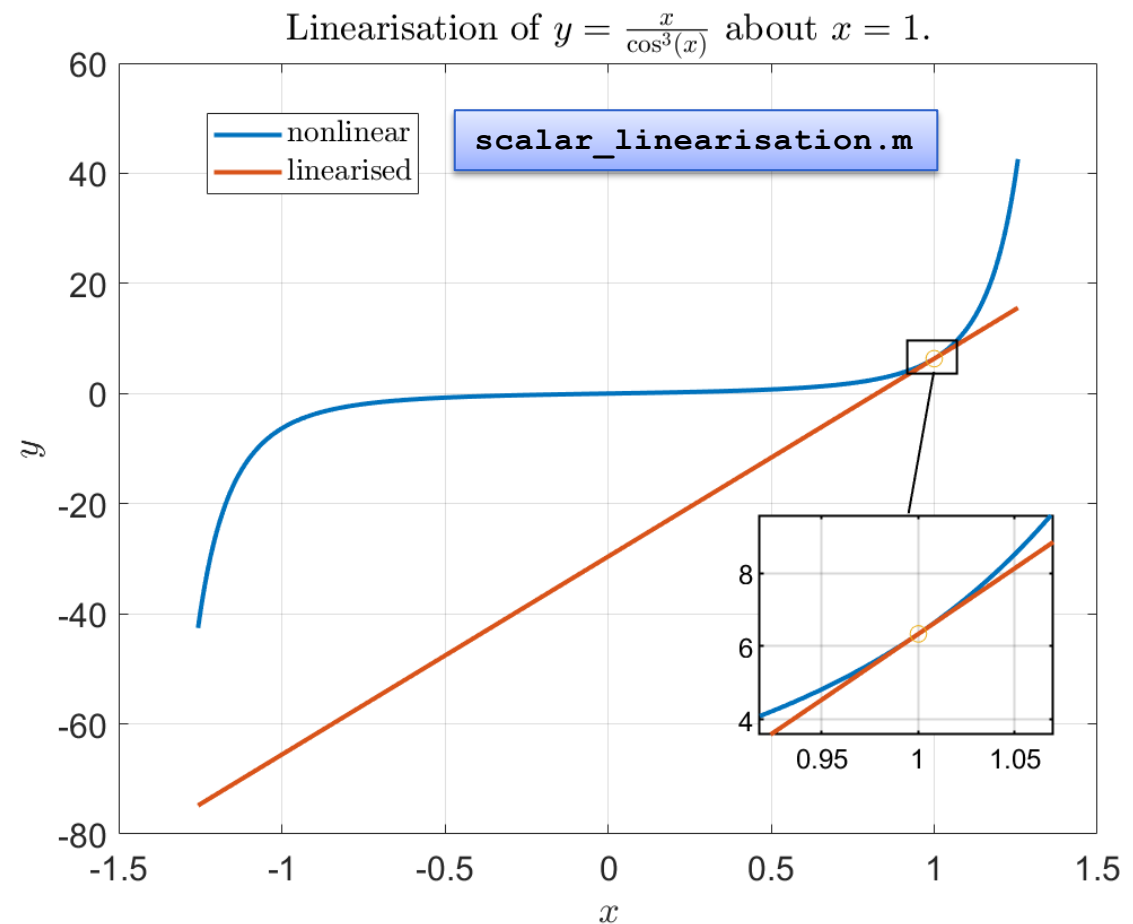
- If the function $f(x)$ is nonlinear, its approximation will be linear in a neighbourhood of x_0

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$$\underbrace{f(x) - f(x_0)}_{\Delta f} \approx f'(x_0) \underbrace{(x - x_0)}_{\Delta x}$$

$$\Delta f \approx f'(x_0) \Delta x \rightarrow \text{linear!}$$

- Small deviations from the operating point x_0 will make the system behave almost as a linear!



Linearisation of nonlinear systems



$$y^{(n)} = \frac{d^n y}{dt^n}$$



- A nonlinear ODE $y^{(n)} = f(y, \dot{y}, \ddot{y}, \dots, y^{(n-1)}, u, \dot{u}, \ddot{u}, \dots, u^{(p)})$ $n \geq p$ depends on more than one variable.
- The Taylor expansion of a **multivariable** function $f(x, y, z)$ around the operating point (x_0, y_0, z_0) is given by

$$f(x, y, z) \approx f(x_0, y_0, z_0) + \frac{\partial f}{\partial x}(x_0, y_0, z_0) \cdot (x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0, z_0) \cdot (y - y_0) + \frac{\partial f}{\partial z}(x_0, y_0, z_0) \cdot (z - z_0)$$

Derivative of $f(x, y, z)$
considering everything
constant except for x

Derivative of $f(x, y, z)$
considering everything
constant except for y

Derivative of $f(x, y, z)$
considering everything
constant except for z

Linearisation of nonlinear systems

Example 1: Linearise the function $f(x, y, z) = x^2 + xy + z \sin(y)$ about the point $(x_0, y_0, z_0) = (2, 0, 2)$

$$f(x, y, z) \approx f(x_0, y_0, z_0) + \boxed{\frac{\partial f}{\partial x}(x_0, y_0, z_0)} \cdot (x - x_0) + \boxed{\frac{\partial f}{\partial y}(x_0, y_0, z_0)} \cdot (y - y_0) + \boxed{\frac{\partial f}{\partial z}(x_0, y_0, z_0)} \cdot (z - z_0)$$

$$\frac{\partial f}{\partial x}(x, y, z) = 2x + y$$

$$\xrightarrow{(2,0,2)} \frac{\partial f}{\partial x}(2, 0, 2) = 4$$

$$\frac{\partial f}{\partial y}(x, y, z) = x + z \cos(y)$$

$$\frac{\partial f}{\partial y}(2, 0, 2) = 4$$

$$\frac{\partial f}{\partial z}(x, y, z) = \sin(y)$$

$$\frac{\partial f}{\partial z}(2, 0, 2) = 0$$

$$\begin{aligned} f(x, y, z) &= f(2, 0, 2) + \frac{\partial f}{\partial x}(2, 0, 2) \cdot (x - 2) + \frac{\partial f}{\partial y}(2, 0, 2) \cdot (y - 0) \\ &+ \frac{\partial f}{\partial z}(2, 0, 2) \cdot (z - 2) + \dots \approx \boxed{4 + 4(x - 2) + 4y} \end{aligned}$$

Valid only close to (2,0,2)

Linearisation of nonlinear systems

Example 2: Linearise the function $z = f(x, y) = xy$ about the point $(x_0, y_0) = (6, 11)$. Evaluate the linearised equation at the point $(x_1, y_1) = (5, 10)$ and compare the result with the one from the nonlinear equation.

$$f(x, y) \approx f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0) \cdot (x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0) \cdot (y - y_0)$$

$$\frac{\partial f}{\partial x}(x, y) = y \Rightarrow \frac{\partial f}{\partial x}(6, 11) = 11 \qquad \frac{\partial f}{\partial y}(x, y) = x \Rightarrow \frac{\partial f}{\partial y}(6, 11) = 6$$

Nonlinear equation at $(5, 10)$: $z = xy = 50$

Linearised equation at $(5, 10)$: $z \approx 66 + 11(x - 6) + 6(y - 11) = 11 \cdot 5 + 6 \cdot 10 - 66 = 49$

The exact value of $z = xy = 50 \rightarrow$ **the error is 1 (2%)**

Linearisation of nonlinear systems

Given a nonlinear ODE $y^{(n)} = f(y, \dot{y}, \ddot{y}, \dots, y^{(n-1)}, u, \dot{u}, \ddot{u}, \dots, u^{(p)})$, $n \geq p$ we will expand considering **all** the variables on the right-hand-side of the equation.

Example 3: Linearise the system $\ddot{y} = \underbrace{-5\dot{y} - 2\dot{u}^2 y^3 + 3u + \sin(u^2 + 1) + 25}_{f(y, \dot{y}, u, \dot{u})}$ about the point $(y_0, \dot{y}_0, u_0, \dot{u}_0) = (7, 0, 0.217, 0)$.

$$\frac{\partial f}{\partial y} = -6\dot{u}^2 y^2 y = 0 \quad \frac{\partial f}{\partial \dot{y}} = -5 \quad \frac{\partial f}{\partial u} = 3 + 2u \cos(u^2 + 1) = 3.434 \quad \frac{\partial f}{\partial \dot{u}} = -4\dot{u} y^3 = 0$$

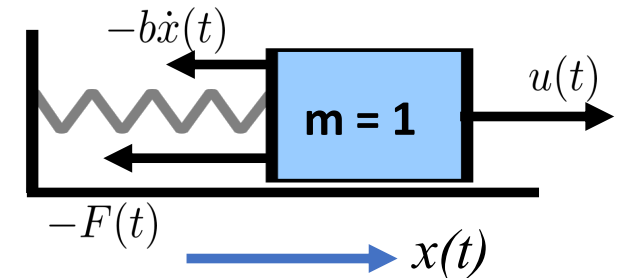
$$\ddot{y} \approx f(y_0, \dot{y}_0, u_0, \dot{u}_0) - 5\dot{y} + 3.434(u - 0.217) \Leftrightarrow \underbrace{\ddot{y} - f(y_0, \dot{y}_0, u_0, \dot{u}_0)}_{\Delta \ddot{y}} \approx \underbrace{-5(\dot{y} - 0)}_{\Delta \dot{y}} + \underbrace{3.434(u - 0.217)}_{\Delta u} \Rightarrow$$

$$\Delta \ddot{y} \approx -5\Delta \dot{y} + 3.434\Delta u \xrightarrow{\mathcal{L}} (s^2 + 5s)\Delta Y(s) = 3.434\Delta U(s) \Leftrightarrow \boxed{G(s) \triangleq \frac{\Delta Y(s)}{\Delta U(s)} = \frac{3.434}{s^2 + 5s}}$$

Linearisation of nonlinear systems

Example 4: Linearise the nonlinear mass-spring-damper system $\ddot{x} = \underbrace{u - kx^2 - b\dot{x}}_{f(x,\dot{x},u)}$ about the point $(x_0, \dot{x}_0, u_0)$ and find the transfer function $G(s) \triangleq \frac{\Delta X(s)}{\Delta U(s)}$.

$$\frac{\partial f}{\partial x}(x_0, \dot{x}_0, u_0) = -2kx_0, \quad \frac{\partial f}{\partial \dot{x}}(x_0, \dot{x}_0, u_0) = -b, \quad \frac{\partial f}{\partial u}(x_0, \dot{x}_0, u_0) = 1$$



$$f(x, \dot{x}, u) \approx f(x_0, \dot{x}_0, u_0) + \frac{\partial f}{\partial x}(x_0, \dot{x}_0, u_0) \cdot (x - x_0) + \frac{\partial f}{\partial \dot{x}}(x_0, \dot{x}_0, u_0) \cdot (\dot{x} - \dot{x}_0) + \frac{\partial f}{\partial u}(x_0, \dot{x}_0, u_0) \cdot (u - u_0) \Leftrightarrow$$

$$\underbrace{f(x, \dot{x}, u) - f(x_0, \dot{x}_0, u_0)}_{\Delta \ddot{x}} \approx \underbrace{-2kx_0}_{\Delta x} \underbrace{(x - x_0)}_{\Delta x} - \underbrace{b}_{\Delta \dot{x}} \underbrace{(\dot{x} - \dot{x}_0)}_{\Delta \dot{x}} + \underbrace{(u - u_0)}_{\Delta u} \Leftrightarrow \boxed{\Delta \ddot{x} \approx -2kx_0 \Delta x - b \Delta \dot{x} + \Delta u}$$

No nonlinearities!

$$\xrightarrow{\mathcal{L}} s^2 \Delta X(s) \approx -2kx_0 \Delta X(s) - bs \Delta X(s) + U(s) \Rightarrow \boxed{G(s) \triangleq \frac{\Delta X(s)}{\Delta U(s)} \approx \frac{1}{s^2 + bs + 2kx_0}}$$

Linearisation of nonlinear systems

Example 5: Linearise the inverted pendulum system about the point

$(\theta_0, \dot{\theta}_0, F_0) = (0, 0.1, 1)$ and find the transfer function $G(s) \triangleq \frac{\Delta\Theta(s)}{\Delta F(s)}$

from input force F to tilt angle θ . Newton's 2nd law for rotational motion:

$$\ddot{\theta} = \underbrace{\frac{1}{mL} F \cos(\theta) - \frac{g}{L} \sin(\theta) - \frac{b}{mL} \cos(\theta) \dot{\theta}}_{f(\theta, \dot{\theta}, F)}$$

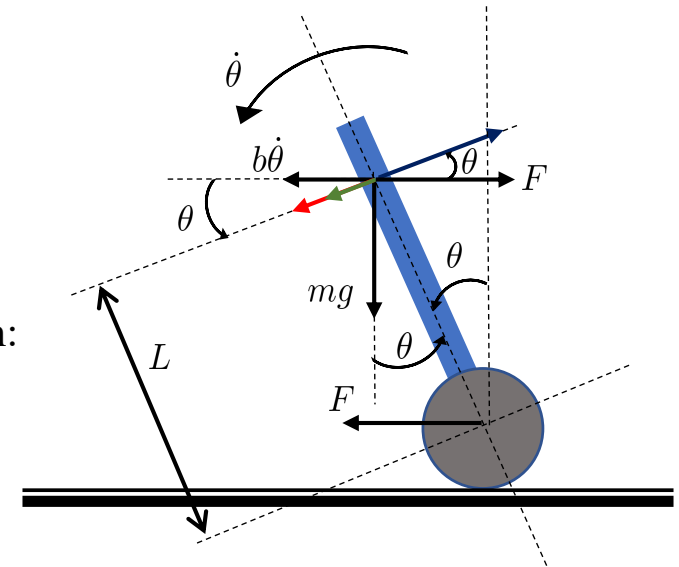
$$\Delta\ddot{\theta} = \underbrace{\left(\frac{-1}{mL} F_0 \sin(\theta_0) - \frac{g}{L} \cos(\theta_0) + \frac{b}{mL} \sin(\theta_0) \dot{\theta}_0 \right)}_{\frac{\partial f}{\partial \theta}(0, 0.1, 1) = -\frac{g}{L}} \Delta\theta + \underbrace{\frac{-b}{mL} \cos(\theta_0)}_{\frac{\partial f}{\partial \dot{\theta}}(0, 0.1, 1) = -\frac{b}{mL}} \Delta\dot{\theta} + \underbrace{\frac{1}{mL} \cos(\theta_0)}_{\frac{\partial f}{\partial F}(0, 0.1, 1) = \frac{1}{mL}} \Delta F \Leftrightarrow$$

$$\Delta\ddot{\theta} = -\frac{g}{L} \Delta\theta - \frac{b}{mL} \Delta\dot{\theta} + \frac{1}{mL} \Delta F$$

No nonlinearities!

$$\xrightarrow{\mathcal{L}} s^2 \Delta\Theta(s) \approx \frac{-g}{L} \Delta\Theta(s) - \frac{b}{mL} s \Delta\Theta(s) + \frac{1}{mL} \Delta F(s) \Rightarrow$$

$$G(s) \triangleq \frac{\Delta\Theta(s)}{\Delta F(s)} \approx \frac{1}{mLs^2 + bs + mg}$$



Linearisation of nonlinear systems - Cookbook

Given a nonlinear ODE $y^{(n)} = f(y, \dot{y}, \ddot{y}, \dots, y^{(n-1)}, u, \dot{u}, \ddot{u}, \dots, u^{(p)}), n \geq p$ and an operating point $p_0 = (y_0, \dot{y}_0, \ddot{y}_0, \dots, y_0^{(n-1)}, u_0, \dot{u}_0, \ddot{u}_0, \dots, u_0^{(p)})$.

Step 1: Calculate the partial derivatives of f with respect to **all** the arguments, e.g. $\frac{\partial f}{\partial y}, \frac{\partial f}{\partial \dot{y}}, \dots, \frac{\partial f}{\partial u}, \frac{\partial f}{\partial \dot{u}}, \dots$

Step 2: Substitute the values of the operating point into the expressions for the partial derivatives.

Step 3: Express the dynamics of the deviations from the operating point as

$$\Delta y^{(n)} \approx \underbrace{\frac{\partial f}{\partial y} \Big|_{p_0}}_{a_0} \Delta y + \underbrace{\frac{\partial f}{\partial \dot{y}} \Big|_{p_0}}_{a_1} \Delta \dot{y} + \underbrace{\frac{\partial f}{\partial y^{(n-1)}} \Big|_{p_0}}_{a_{n-1}} \Delta y^{(n-1)} + \dots + \underbrace{\frac{\partial f}{\partial u} \Big|_{p_0}}_{b_0} \Delta u + \underbrace{\frac{\partial f}{\partial \dot{u}} \Big|_{p_0}}_{b_1} \Delta \dot{u} + \dots + \underbrace{\frac{\partial f}{\partial u^{(p)}} \Big|_{p_0}}_{b_p} \Delta u^{(p)}$$

Step 4: Apply the Laplace transform and factorise the powers of each signal

$$\xrightarrow{\mathcal{L}} (s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0)\Delta Y(s) = (b_p s^p + \dots + b_1s + b_0)U(s)$$

Step 5: Solve for $\Delta Y(s)/\Delta U(s) \rightarrow G(s) \triangleq \frac{\Delta Y(s)}{\Delta U(s)} = \frac{b_p s^p + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$



$$y^{(n)} = \frac{d^n y}{dt^n}$$

Today you learned

- Types of modelling for dynamical systems
- Black-box modelling with Matlab
- Conversion from ODEs to Laplace for finding transfer functions
- Linearisation of nonlinear systems

Next lecture 6

Frequency analysis

- Bode plot - 1st and 2nd order
- Stability
- Stability margins

Today's Exercises

Group work: Black Box Modeling for REGBOT

Home alone: Extra exercises (pdf file on Block diagram, TF)